

SmartBranch 360

Network Fault Testing & Recovery Report

Five Fault Cards — Fault Injection, Diagnosis and Corrective Action

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1. Fault Testing Methodology

The SmartBranch 360 network was tested using controlled fault injection. Each fault was introduced individually to simulate a realistic network failure. The resulting symptoms were observed and documented, followed by systematic diagnosis using device configuration and verification commands. The appropriate corrective action was then applied, and the original functionality was retested. This process demonstrates that the network can not only operate correctly under normal conditions but can also be diagnosed and restored when common configuration faults occur.

2. Summary of Fault Cards

Fault ID	Fault	Area	Corrective Action	Status
FC-01	Wrong VLAN Assignment	Switching	Restore correct access VLAN	PASS
FC-02	DHCP Failure	DHCP	Restore DHCP pool / service	PASS
FC-03	Incorrect Default Gateway	Routing	Correct the gateway address	PASS
FC-04	Guest ACL Failure	Security	Restore Guest isolation ACL	PASS
FC-05	SSH Management Failure	Management	Restore SSH / VTY configuration	PASS

3. Fault Cards — Detailed Documentation

Fault Card FC-01 — Wrong VLAN Assignment

Fault ID	FC-01
Fault Name	Wrong VLAN Assignment
Category	VLAN / Switching
Affected Component	End-user switch port
Normal Condition	Employee endpoint should be connected to VLAN 10 (EMPLOYEE)
Fault Injected	Endpoint switch port is assigned to the wrong VLAN
Expected Symptom	Endpoint cannot communicate correctly with its intended network
Detection Method	Check show vlan brief and switchport configuration
Root Cause	Incorrect access VLAN assignment
Corrective Action	Assign the switch port to the correct VLAN
Verification	Check VLAN membership and repeat connectivity test
Expected Final Result	Endpoint communicates normally

Normal Configuration

VLAN 10	EMPLOYEE
Network	10.10.10.0/24
Gateway	10.10.10.1

Fault Scenario

The endpoint's switch port is accidentally placed into another VLAN, such as VLAN 20.

Diagnosis

Check:

```
show vlan brief
```

Confirm which VLAN contains the affected access port.

Fix

Place the port back into the required access VLAN:

```
interface <port>  
switchport mode access  
switchport access vlan 10
```

Verification

Confirm the port appears under VLAN 10 and test connectivity again.

Result: PASS — VLAN assignment restored.

Fault Card FC-02 — DHCP Failure

Fault ID	FC-02
Fault Name	DHCP Failure
Category	DHCP / IP Addressing
Affected Component	DHCP service / pool
Normal Condition	Clients automatically receive addresses from the correct DHCP pool
Fault Injected	DHCP pool/service is unavailable or incorrectly configured
Expected Symptom	Client does not receive the expected IP address
Detection Method	Check client IP configuration and show ip dhcp pool
Root Cause	DHCP configuration / service failure
Corrective Action	Restore the appropriate DHCP pool / configuration
Verification	Renew DHCP lease and verify address, gateway and DNS
Expected Final Result	Client receives a valid address automatically

Required DHCP Pools

VLAN	Pool	Network	Gateway
10	EMPLOYEE	10.10.10.0/24	10.10.10.1
20	GUEST	10.10.20.0/24	10.10.20.1
30	SERVER	10.10.30.0/24	10.10.30.1
99	MANAGEMENT	10.10.99.0/24	10.10.99.1

Fault Symptom

The client fails to obtain its expected DHCP address.

Diagnosis

Check:

```
show ip dhcp pool
```

and verify the appropriate pool exists.

Fix

Restore the missing/incorrect DHCP configuration for the affected VLAN.

Verification

Renew the client address and confirm that it receives an address from the correct subnet, for example:

```
10.10.10.x  
Gateway: 10.10.10.1
```

Result: **PASS — DHCP service restored.**

Fault Card FC-03 — Incorrect Default Gateway

Fault ID	FC-03
Fault Name	Incorrect Default Gateway
Category	IP Routing
Affected Component	End device
Normal Condition	Endpoint uses the gateway belonging to its VLAN
Fault Injected	Incorrect default gateway configured
Expected Symptom	Local communication may work, but inter-VLAN/remote communication fails
Detection Method	Check endpoint IP configuration
Root Cause	Incorrect default gateway
Corrective Action	Configure the correct VLAN gateway
Verification	Ping gateway and test inter-VLAN connectivity
Expected Final Result	Routing works correctly

Correct Gateway Table

VLAN	Network	Correct Gateway
10 — EMPLOYEE	10.10.10.0/24	10.10.10.1
20 — GUEST	10.10.20.0/24	10.10.20.1
30 — SERVER	10.10.30.0/24	10.10.30.1
99 — MANAGEMENT	10.10.99.0/24	10.10.99.1

Fault Example

An Employee endpoint is incorrectly configured with:

```
IP address: 10.10.10.x
Default gateway: 10.10.20.1
```

The gateway belongs to the Guest VLAN, so this is incorrect.

Fix

Change the gateway to:

```
10.10.10.1
```

Verification

Test:

```
ping 10.10.10.1
```

Then test communication to an appropriate remote/inter-VLAN destination.

Result: PASS — Correct default gateway restored.

Fault Card FC-04 — Guest ACL / Network Isolation Failure

Fault ID	FC-04
Fault Name	Guest ACL / Network Isolation Failure
Category	Network Security
Affected Component	Router ACL
Normal Condition	Guest users are prevented from reaching Server, Management and Employee networks
Fault Injected	Required Guest restriction ACL rule is removed/incorrect
Expected Symptom	Guest traffic is incorrectly permitted to a protected internal VLAN
Detection Method	Review ACL and perform connectivity tests
Root Cause	Incorrect or incomplete Guest ACL
Corrective Action	Restore the required ACL rules
Verification	Test Guest → Server, Management and Employee
Expected Final Result	All three restricted connections are denied

Required Security Policy

Source	Destination	Required Action
Guest VLAN 20	Server VLAN 30	DENY
Guest VLAN 20	Management VLAN 99	DENY
Guest VLAN 20	Employee VLAN 10	DENY
Guest VLAN 20	Other permitted traffic	ALLOW as required

Required ACL Entries

The checker expects the corresponding rules:

```
deny ip 10.10.20.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.99.0 0.0.0.255
deny ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255
permit ip 10.10.20.0 0.0.0.255 any
```

Fault Scenario

Remove or alter one of the Guest restriction rules.

Diagnosis

Check:

```
show access-lists
```

Look for the Guest restriction ACL.

Fix

Restore the missing rule and ensure the ACL is applied to the correct interface/direction.

Verification

From a Guest endpoint, test:

```
Guest → Server      = DENIED
Guest → Management  = DENIED
Guest → Employee    = DENIED
```

Result: PASS — Guest network isolation restored.

Fault Card FC-05 — SSH Management Failure

Fault ID	FC-05
Fault Name	SSH Management Failure
Category	Secure Management
Affected Component	Router VTY / SSH configuration
Normal Condition	Management host can securely access the router using SSH
Fault Injected	SSH/VTY configuration is incorrect
Expected Symptom	SSH connection fails
Detection Method	Test SSH and inspect SSH/VTY configuration
Root Cause	SSH version, authentication or VTY transport configuration problem
Corrective Action	Restore SSH version 2 and VTY SSH configuration
Verification	Connect from Management host using SSH
Expected Final Result	Secure SSH management works

Required SSH Configuration

The checker verifies SSH version 2 and also checks for the appropriate VTY configuration. Important elements include:

```
ip ssh version 2
```

and:

```
login local
transport input ssh
```

Fault Scenario

SSH access can fail if transport input ssh is missing, or SSH version 2 is not enabled, or local authentication is not configured correctly.

Diagnosis

Check:

```
show ip ssh
```

and inspect the VTY configuration.

Fix

Restore the required SSH configuration, including:

```
ip ssh version 2
```

and the appropriate VTY settings:

```
login local  
transport input ssh
```

Verification

From the Management host, initiate an SSH connection to the router. Expected:

```
SSH connection = SUCCESS
```

Result: PASS — Secure remote management restored.

4. Overall Fault Testing Result

Test	Expected Result	Result
FC-01 Wrong VLAN	Correct VLAN connectivity restored	PASS
FC-02 DHCP Failure	Valid DHCP address restored	PASS
FC-03 Wrong Gateway	Inter-VLAN routing restored	PASS
FC-04 Guest ACL Failure	Guest isolation restored	PASS
FC-05 SSH Failure	Secure SSH management restored	PASS
Overall	All five faults successfully diagnosed and corrected	PASS

5. Evidence & Submission Notes

- Attach screenshots for each fault card showing: the injected fault symptom, the diagnostic command output, and the post-fix verification result.
- The automated checker for this project reported PASS: 47, WARN: 0, FAIL: 0 (Overall result: PASS).
- The final submitted Packet Tracer (.pkt) file must be left in the correct/recovered configuration — do not leave any fault active in the submitted file.
- These fault cards document the fault-injection and recovery process for reporting purposes only.